

JEE MAIN REVISION MASTER: ALCOHOL, PHENOLS AND ETHERS

Subject: Chemistry | Target Scope: JEE Main Previous Years | Empirical Base: 32 Analyzed PYQs

1. High-Yield Concept Priority Matrix

Priority Tier	Concept Name	Frequency	Category
Very High	Williamson Ether Synthesis & Cleavage by HI	17 PYQs	Ethers
Very High	Acidity of Phenols & Distinguishing Tests	15 PYQs	Phenols

2. Master Formula Reference Card

Concept / Topic	Mathematical Formulation & Governing Equations
Williamson Ether Synthesis & Cleavage by HI	$\text{Williamson: } \text{R-O}^- + \text{Na}^+ + \text{R}'\text{-X (must be } 1^\circ) \rightarrow \text{R-O-R}' + \text{NaX}$
Acidity of Phenols & Distinguishing Tests	$\text{Acidity: Picric acid} > \text{p-Nitrophenol} > \text{Phenol} > \text{Water} > \text{Ethanol}, \quad \text{Lucas: } 3^\circ \text{ (turbidity immediately)}, 2^\circ \text{ (5 mins)}, 1^\circ \text{ (no turbidity)}$

3. Deep-Dive Concepts & Linked PYQ Evidence

3.1 Williamson Ether Synthesis & Cleavage by HI (Very High Yield — 17 PYQs)

Core Principle: Alkoxide SN2 displacement on primary halides, and ether cleavage mechanism with concentrated HI.

■ **Caution / JEE Pitfall:** In Williamson synthesis, the alkyl halide MUST be primary (1°). If a tertiary (3°) halide is used, elimination dominates to give an ALKENE instead of an ether!

■ **Speed Trick:** With unsymmetrical ethers + HI: If one alkyl group is tertiary (3°), the iodide attaches to the 3° carbon (via SN1). If both are 1° or 2°, iodide attacks the less hindered carbon (via SN2).

Representative Source PYQs:

- [2026 Shift] Q#1 (MCQ): "The correct IUPAC name of is..." [Ans: N/A]
- [2026 Shift] Q#2 (MCQ): "Glycerol is a trihydric alcohol. It contains ____." [Ans: B]
- [2026 Shift] Q#3 (MCQ): "R-CH₂(2)OH is converted into R-CHO by reacting with ____." [Ans: N/A]

3.2 Acidity of Phenols & Distinguishing Tests (Very High Yield — 15 PYQs)

Core Principle: Phenoxide resonance stabilization, substituent effects on acidity, Reimer-Tiemann and Kolbe-Schmitt reactions, and Lucas test.

■ **Caution / JEE Pitfall:** Electron withdrawing groups ($-\text{NO}_2$, $-\text{CN}$) INCREASE phenol acidity (especially at ortho and para positions), while electron donating groups ($-\text{CH}_3$) decrease acidity.

■ **Speed Trick:** Reimer-Tiemann reaction involves dichlorocarbene ($:\text{CCl}_2$) as an electrophilic intermediate to convert phenol to salicylaldehyde.

Representative Source PYQs:

- **[2025 Shift] Q#3 (MCQ):** "Match the following with their pK_a values Acid pK_a (I) Phenol (a) 16 (II) *p*-Nitrophenol (b) 0.78 (III) Ethyl alcohol (c) 10 (I..." **[Ans: N/A]**
- **[2024 Shift] Q#1 (MCQ):** "8.8 g of monohydric alcohol added to ethyl magnesium iodide in ether liberates 2240cm^3 of ethane at STP. This monohydric alcohol when o..." **[Ans: N/A]**
- **[2023 Shift] Q#2 (MCQ):** "Y in the above reaction is..." **[Ans: N/A]**

4. Benchmark Illustrative PYQs with Master Solutions

Benchmark Problem 1: KCET 2026 (Q#1)

The correct IUPAC name of is

- (A) Tertiary butoxy ethane
- (B) 1,1-Dimethyl-1-ethoxyethane
- (C) 2-ethoxy-2-methyl propane
- (D) Ethoxy tertiary butane

Correct Answer: None

Solution Analysis: In an ether, the larger carbon part is chosen as the parent hydrocarbon, and the smaller part attached through oxygen is named as an alkoxy group. Here, one side of oxygen is an ethyl group, $-\text{C}_2\text{H}_5$. So this part becomes the alkoxy group: ethoxy, $-\text{O}-\text{C}_2\text{H}_5$. The other side is a tert-butyl group, $-\text{C}(\text{CH}_3)_3$. The longest continuous chain in this part has 3 carbon atoms, so the parent hydrocarbon is propane. In the propane chain, the middle carbon is carbon number 2. Both substituents are attached to this carbon. At carbon 2, there is one methyl group, $-\text{CH}_3$, and one etho

Benchmark Problem 2: KCET 2024 (Q#1)

8.8 g of monohydric alcohol added to ethyl magnesium iodide in ether liberates 2240cm^3 of ethane at STP. This monohydric alcohol when oxidised using pyridinium-chloro-chromate, forms a carbonyl compound that answers silver mirror test (Tollen's test). The monohydric alcohol is

- (A) butan-2-ol
- (B) 2,2-dimethyl propan-1-ol
- (C) pentan-2-ol
- (D) 2, 2-dimethyl ethan-1-ol

Correct Answer: None

Solution Analysis: Given that, 8.8 g of monohydric alcohol liberates 2240cm^3 of ethane. So, 88 g of monohydric alcohol will liberate 22400cm^3 (STP) Now, the carbonyl compound that can oxidise Tollens reagent is aldehyde and it is the oxidising product of 1° alcohol. So, among the given options 2, 2-dimethyl propan-1-ol (molecular mass = 88) is the required monohydric alcohol.